

BEGIN

// STEP 1: LOAD DATA FROM ONLINE SOURCE

SET url = transport.json link

SEND HTTP GET request to url

STORE response as trips_data

DISPLAY trips_data

// STEP 2: CONNECT TO FIREBASE

LOAD firebase credentials

INITIALIZE firebase with database URL

DISPLAY "Firebase connected successfully"

// STEP 3: LOAD LOCAL JSON FILE

OPEN transport.json file

READ data into variable "data"

DISPLAY "JSON file loaded"

// STEP 4: UPLOAD DATA TO FIREBASE

SET reference = "trips" in database

FOR each trip IN data DO

 SAVE trip using trip_id as key

END FOR

DISPLAY "Data uploaded successfully"

// STEP 5: RETRIEVE DATA FROM FIREBASE

GET all data from "trips"

IF data exists THEN

 CONVERT to list → trips_list

 DISPLAY number of trips and first trip

ELSE

 SET trips_list = empty

 DISPLAY "No trips found"

END IF

// STEP 6: INITIALIZE SEARCH HISTORY

SET search_history = empty list

// STEP 7: MAIN PROGRAM LOOP

WHILE TRUE DO

 DISPLAY MENU:

 1. Features

 2. Display all trips

3. Add new trip
4. Update trip
5. Delete trip
6. Exit

INPUT user_choice

IF user_choice == 1 THEN

 // FEATURES MENU

 WHILE TRUE DO

 DISPLAY:

- a. Search trips
- b. View search history
- c. Back

 INPUT submenu_choice

 IF submenu_choice == 'a' THEN

 DISPLAY "Searching..."

 SET valid_criteria = [destination, bus_type, trip_id, status]

 REPEAT

 INPUT search_criterion

 UNTIL valid

 DISPLAY available values for that criterion

 INPUT search_value

 IF criterion is trip_id THEN

 CONVERT search_value to integer

 END IF

 SET filtered_trips = empty list

 FOR each trip IN trips_list DO

 IF trip matches criterion AND value THEN

 ADD trip to filtered_trips

 END IF

 END FOR

 IF filtered_trips not empty THEN

 DISPLAY results

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        ADD results to search_history
    ELSE
        DISPLAY "No trips found"
    END IF

ELSE IF submenu_choice == 'b' THEN
    IF search_history empty THEN
        DISPLAY "No history"
    ELSE
        DISPLAY all search_history
    END IF

ELSE IF submenu_choice == 'c' THEN
    BREAK

ELSE
    DISPLAY "Invalid choice"
END IF
END WHILE

ELSE IF user_choice == 2 THEN
    // DISPLAY ALL TRIPS
    IF trips_list not empty THEN
        DISPLAY all trips
    ELSE
        DISPLAY "No trips available"
    END IF

ELSE IF user_choice == 3 THEN
    // ADD NEW TRIP
    CREATE empty new_trip

    INPUT all trip details (id, route, origin, etc.)

    CALCULATE occupancy_rate:
        IF capacity > 0 THEN
            occupancy = passengers / capacity * 100
        ELSE
            occupancy = 0
        END IF

    INPUT fare details (regular, student, senior)

    IF trip_id already exists THEN

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        DISPLAY "Trip already exists"
    ELSE
        SAVE new_trip to Firebase
        DISPLAY "Trip added successfully"
        REFRESH trips_list
    END IF

ELSE IF user_choice == 4 THEN
    // UPDATE TRIP
    INPUT trip_id_to_update
    GET existing_trip from Firebase

    IF trip exists THEN
        DISPLAY current details

        DISPLAY update options:
            a. Status
            b. Fare
            c. Other fields

        INPUT update_choice

        IF 'a' THEN
            UPDATE status
        ELSE IF 'b' THEN
            UPDATE fare values
        ELSE IF 'c' THEN
            UPDATE multiple fields
            RECALCULATE occupancy_rate
        ELSE
            DISPLAY "Invalid choice"
        END IF

        SAVE updates
        REFRESH trips_list

    ELSE
        DISPLAY "Trip not found"
    END IF

ELSE IF user_choice == 5 THEN
    // DELETE TRIP
    INPUT trip_id_to_delete

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    IF trip exists THEN
        DELETE trip
        DISPLAY "Deleted successfully"
        REFRESH trips_list
    ELSE
        DISPLAY "Trip not found"
    END IF

ELSE IF user_choice == 6 THEN
    DISPLAY "Goodbye"
    BREAK LOOP

ELSE
    DISPLAY "Invalid choice"
END IF

END WHILE

END
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